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(54) TAMPER-EVIDENT CONTAINER WITH ROTATING LID

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B65D 17/32 (2006.01)
B65D 17/00 (2006.01)
B65D 55/02 (2006.01)

(52) **U.S. Cl.**
CPC ***B65D 17/32*** (2018.01); ***B65D 17/02*** (2013.01); ***B65D 55/02*** (2013.01)

(58) **Field of Classification Search**
CPC B65D 43/162; B65D 43/16; B65D 43/02; B65D 43/0235; B65D 43/0237; B65D 43/0239; B65D 43/0254; B65D 1/34; B65D 2401/15; B65D 2401/25; B65D 2401/35; B65D 2401/10; B65D 55/06
See application file for complete search history.

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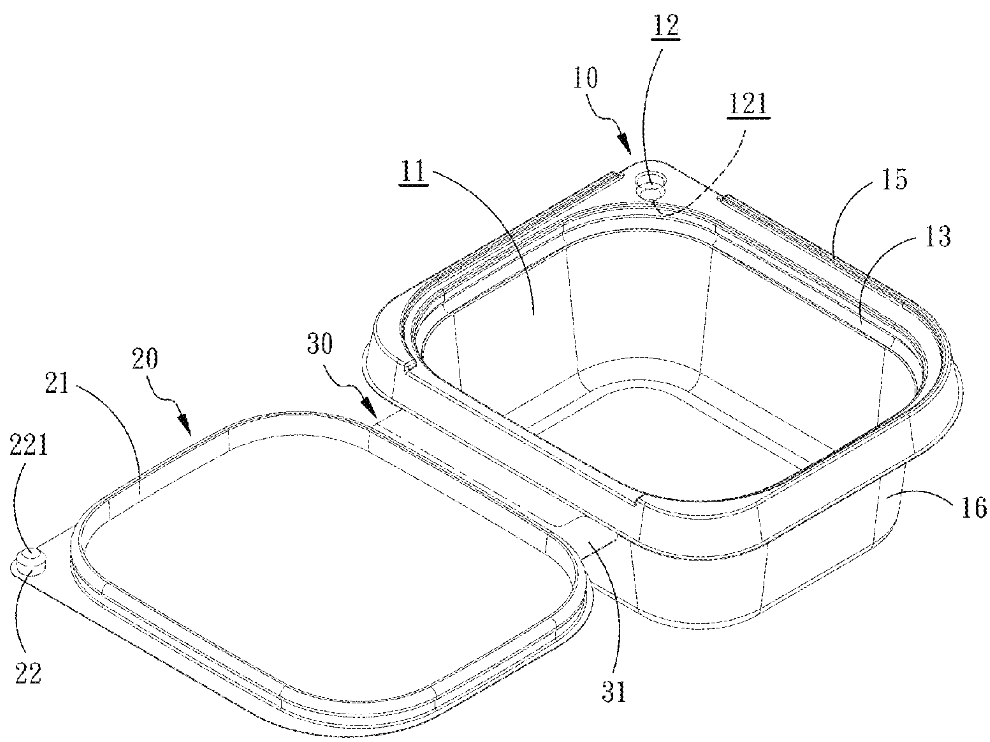
(57) **ABSTRACT**

A tamper-evident container with a rotating lid is disclosed. The tamper-evident container includes a box body, the lid, and a pre-scored connecting portion connecting the box body and the lid. The horizontal end surface of the box body has a first locking portion, and the bottom surface of the lid has a second locking portion that can be detachably engaged with the first locking portion. When the lid is closed with the box body, the container is sealed. By breaking the pre-scored connecting portion, the lid can be horizontally rotated against the first locking portion and the second locking portion, so as to separate the lid from the box body. This structure allows for easy detection of tampering by observing the breakage of the pre-scored connecting portion and enables the container to remain open during use.

16 Claims, 15 Drawing Sheets



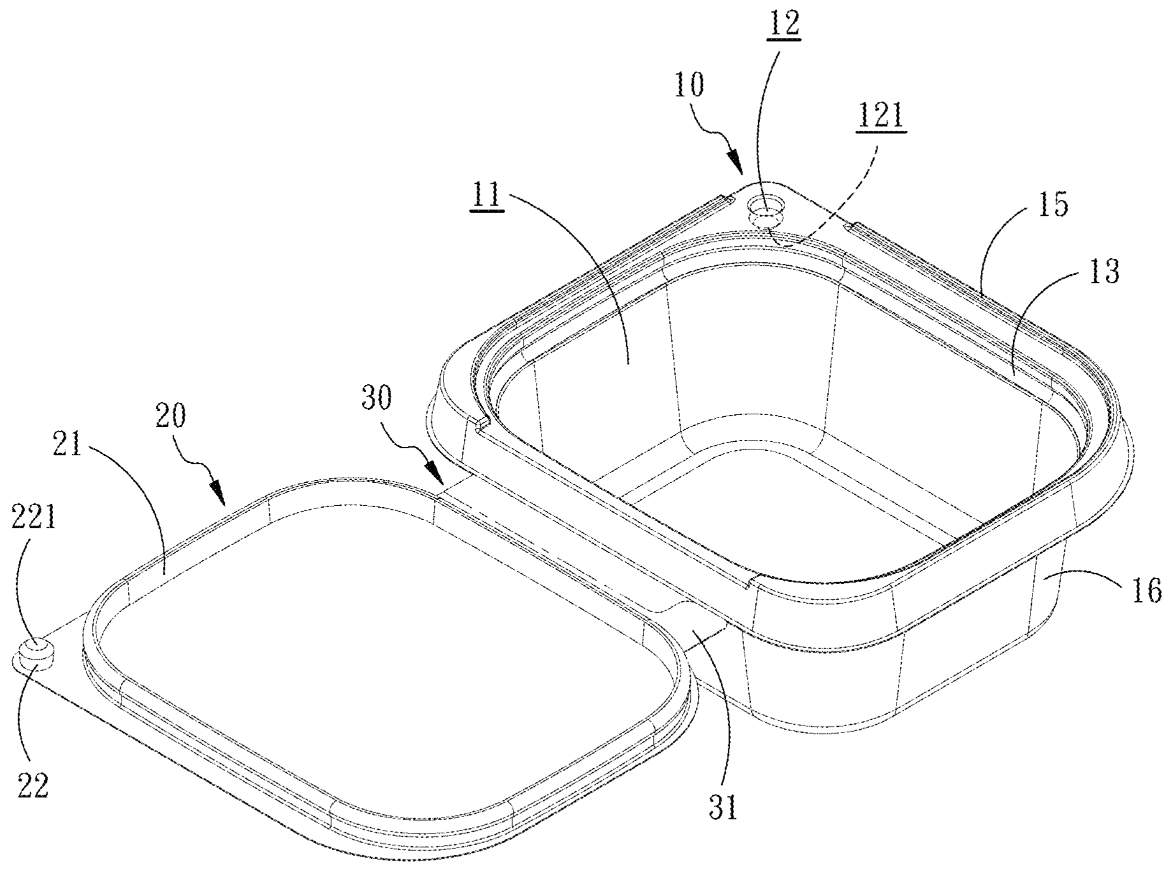


FIG. 1

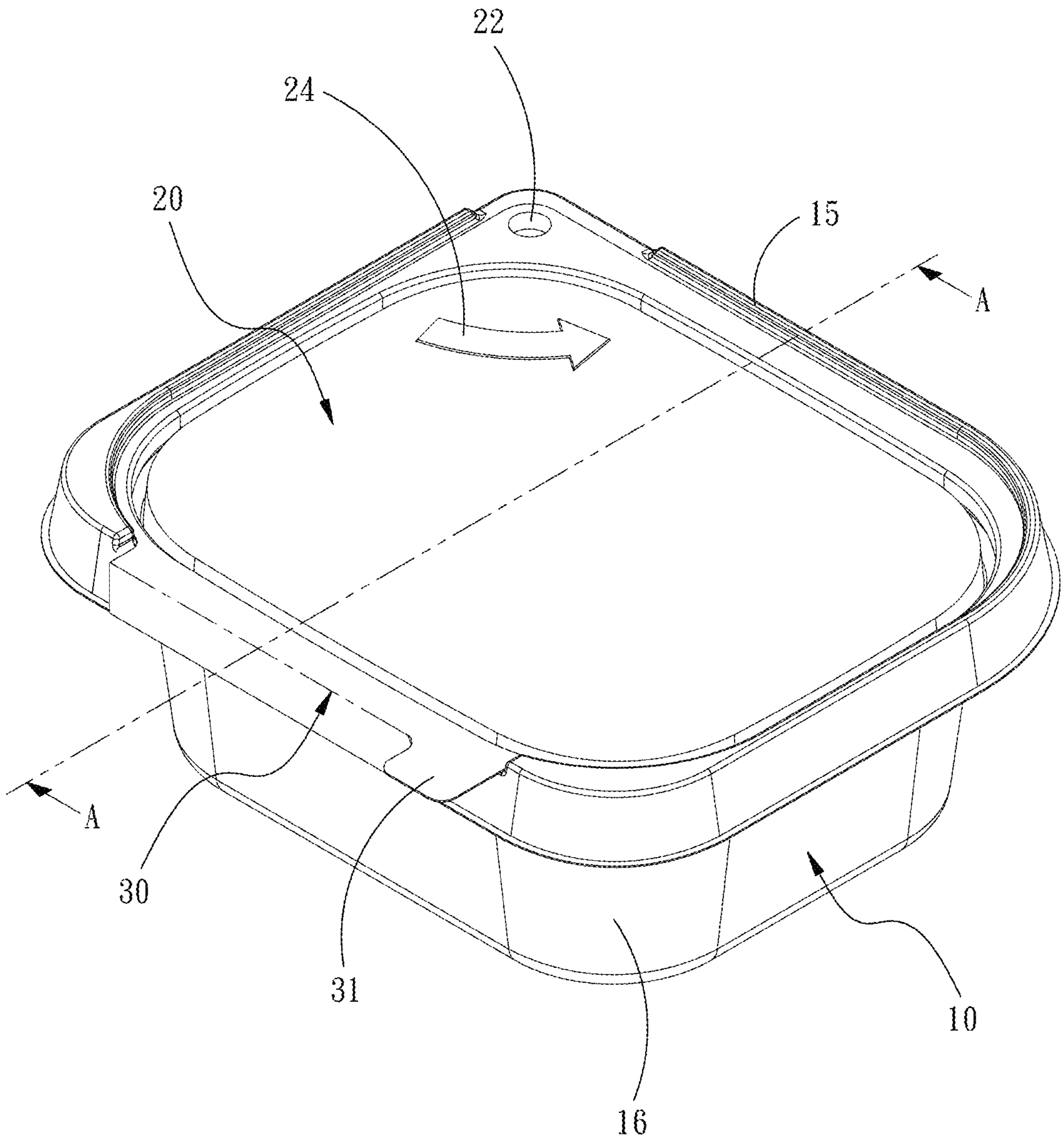


FIG. 2

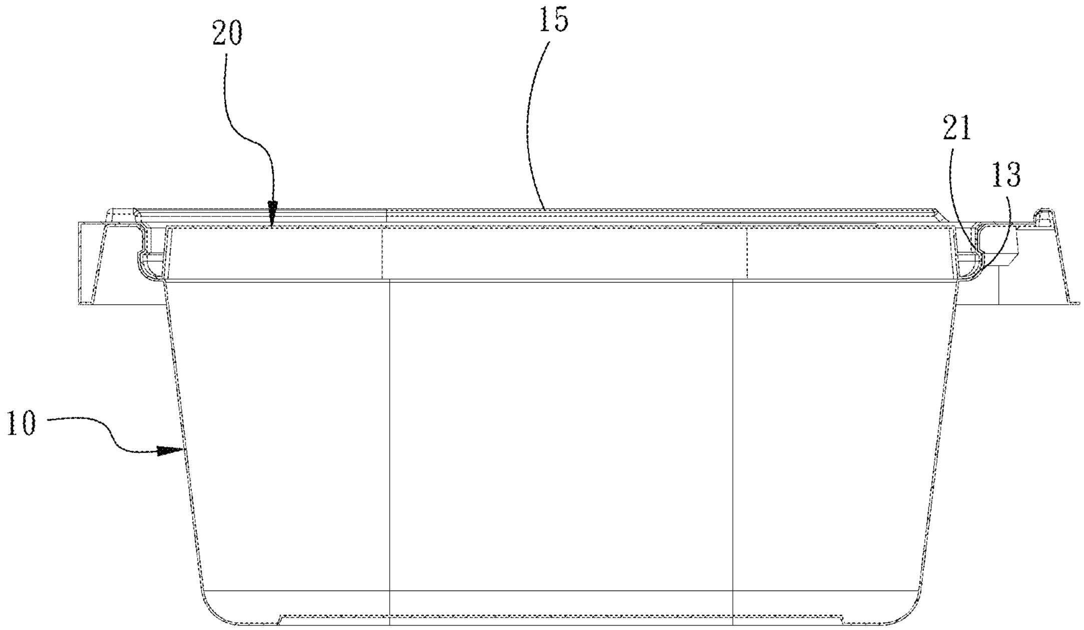


FIG. 3

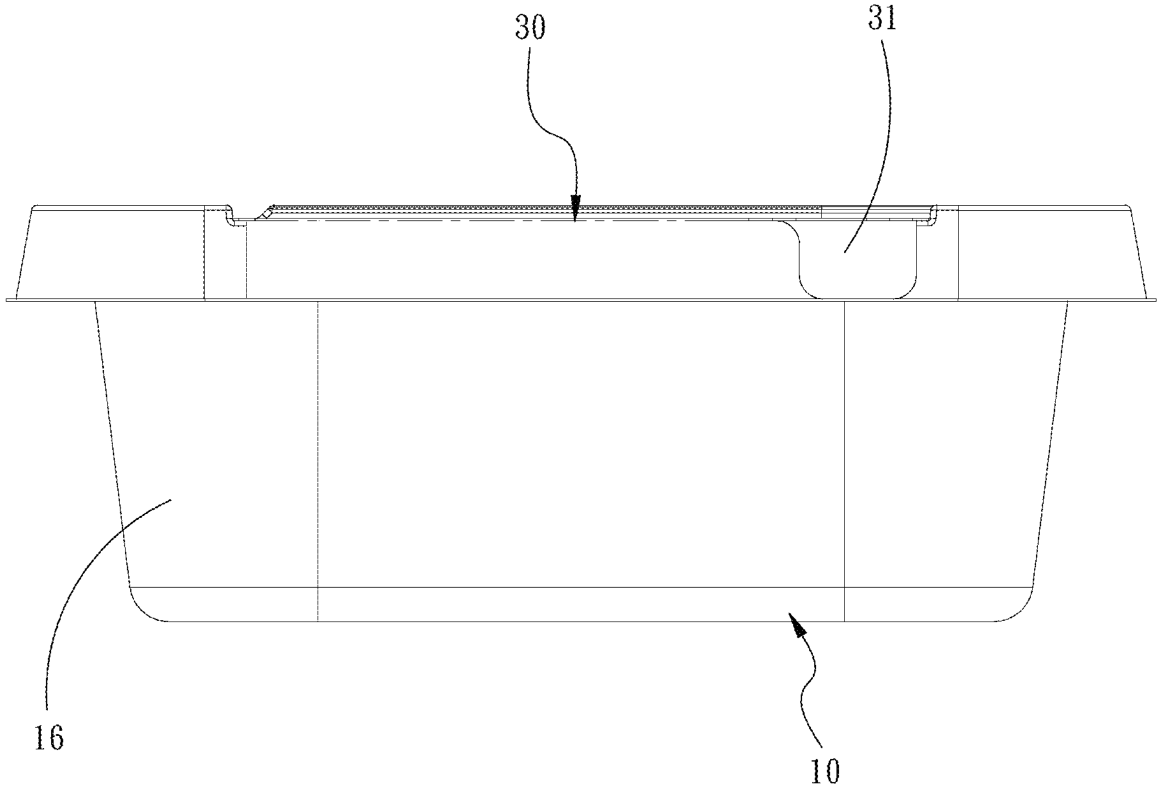


FIG. 4

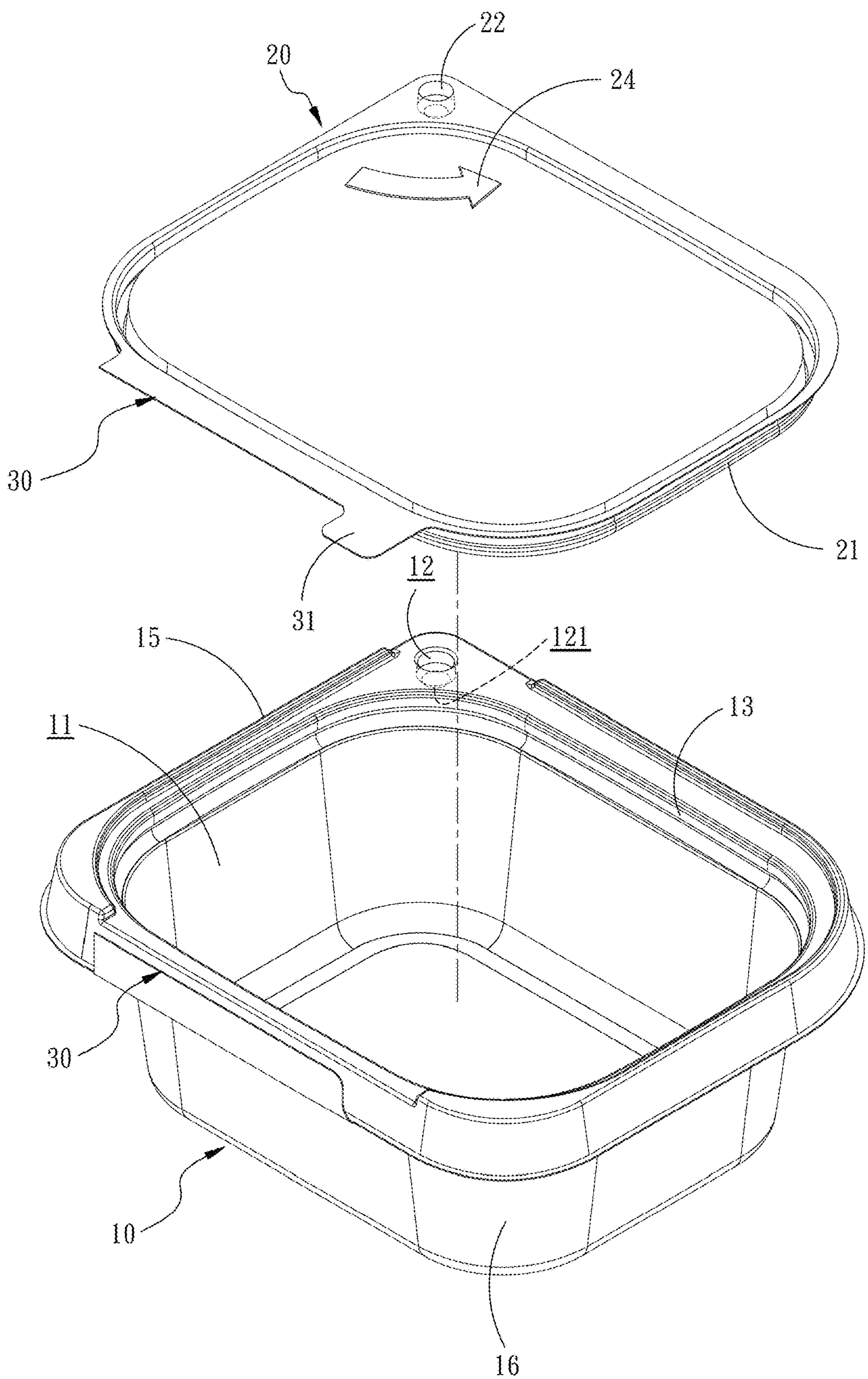


FIG. 5

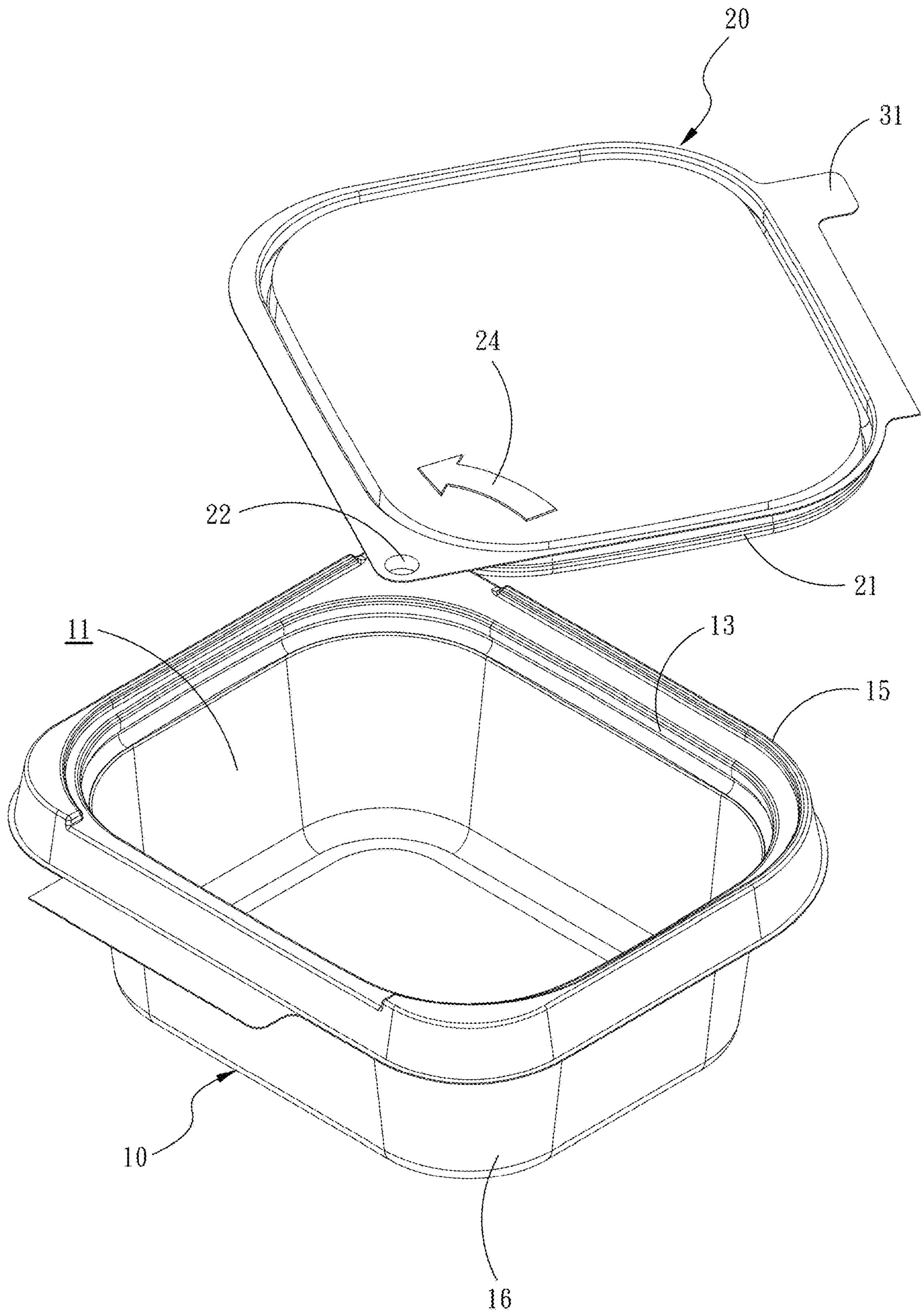


FIG. 6

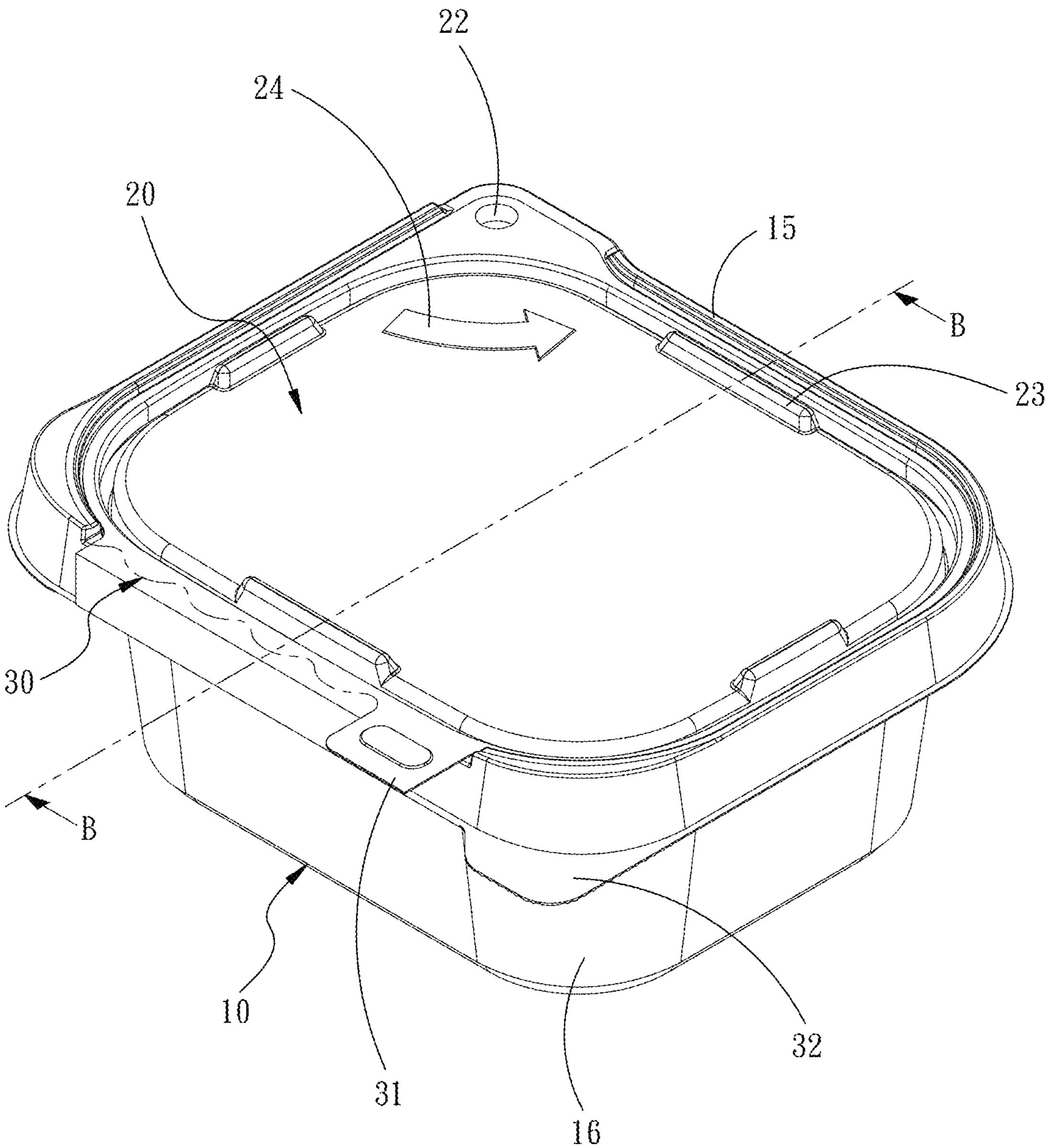


FIG. 7

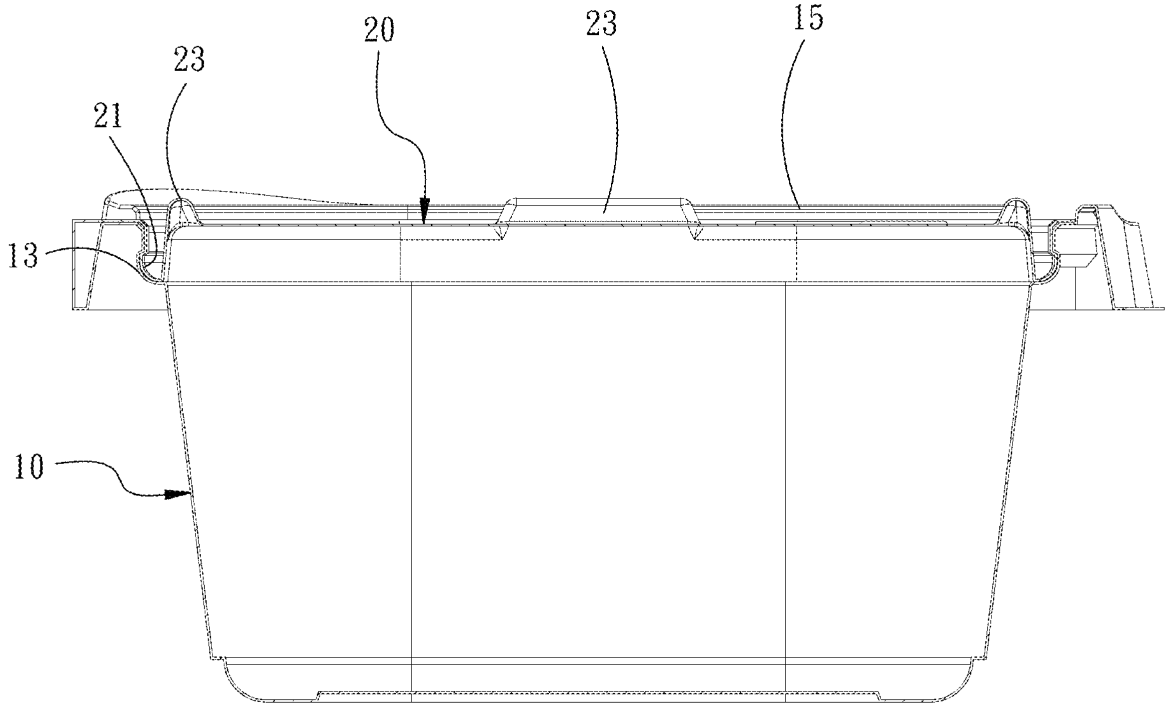


FIG. 8

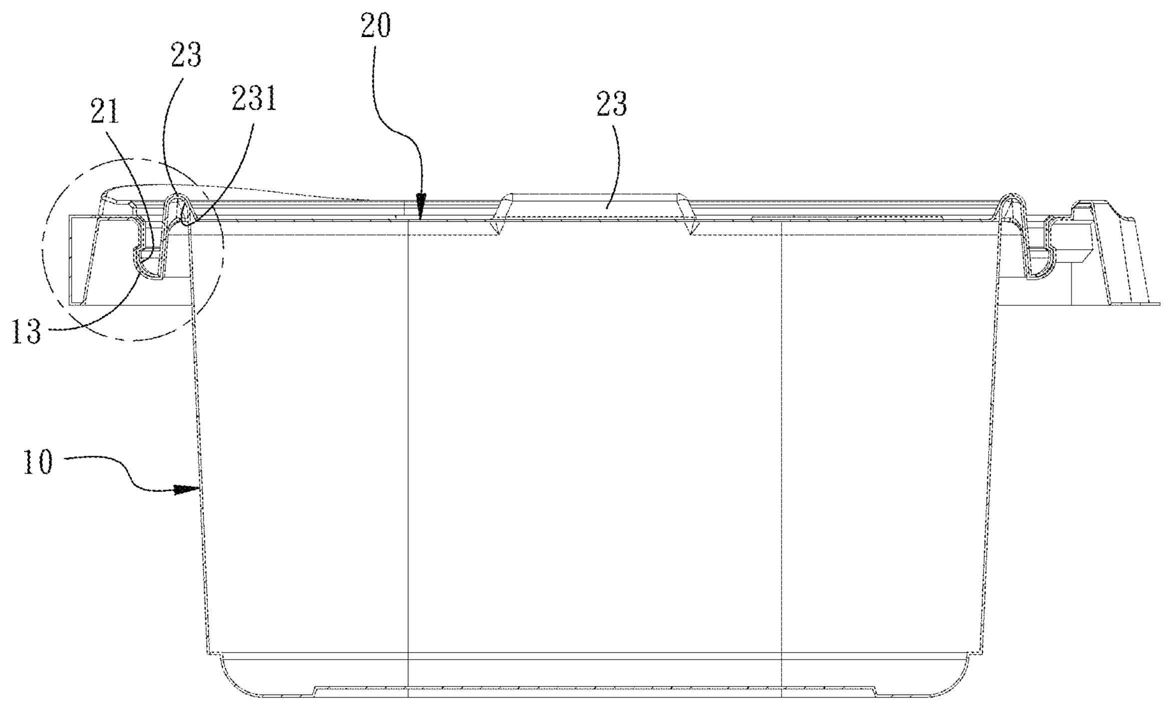


FIG. 9

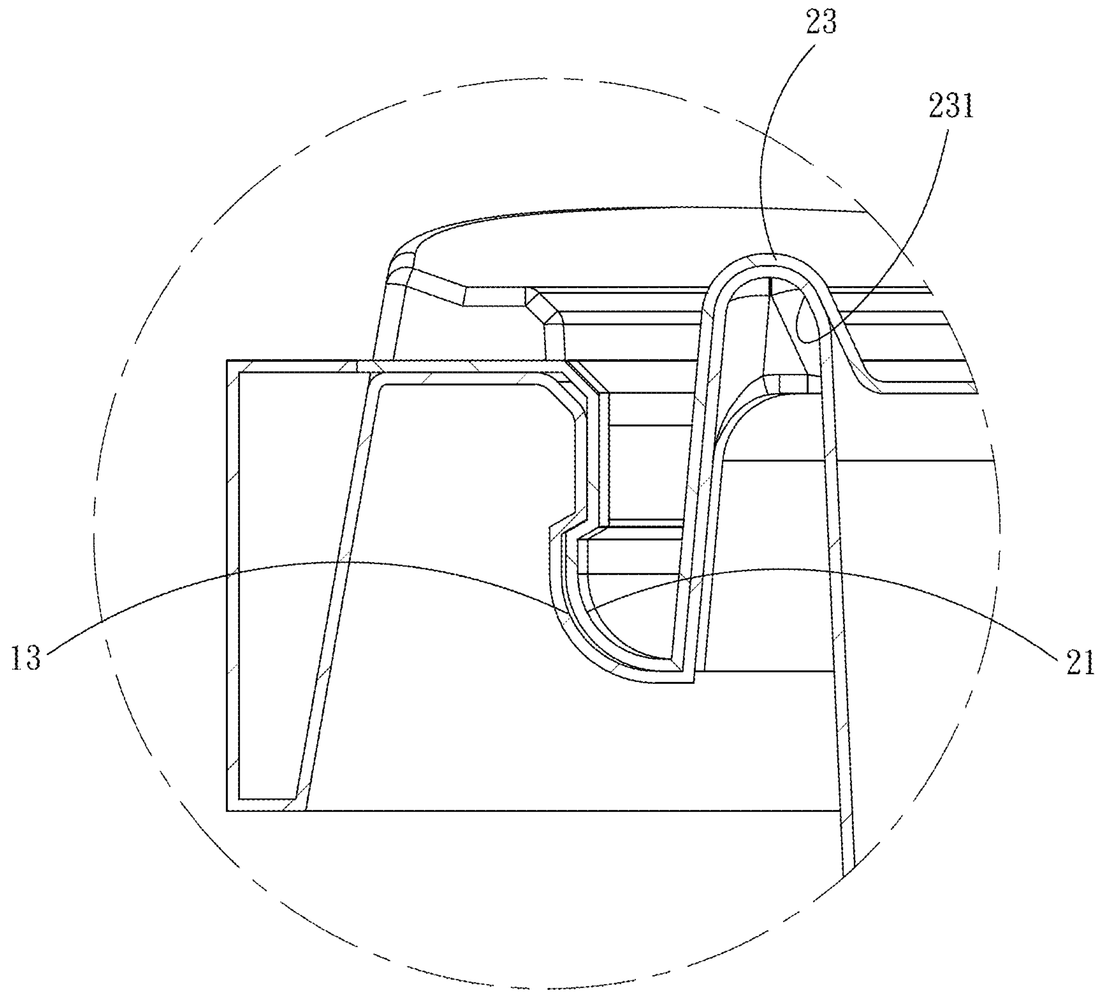


FIG. 10

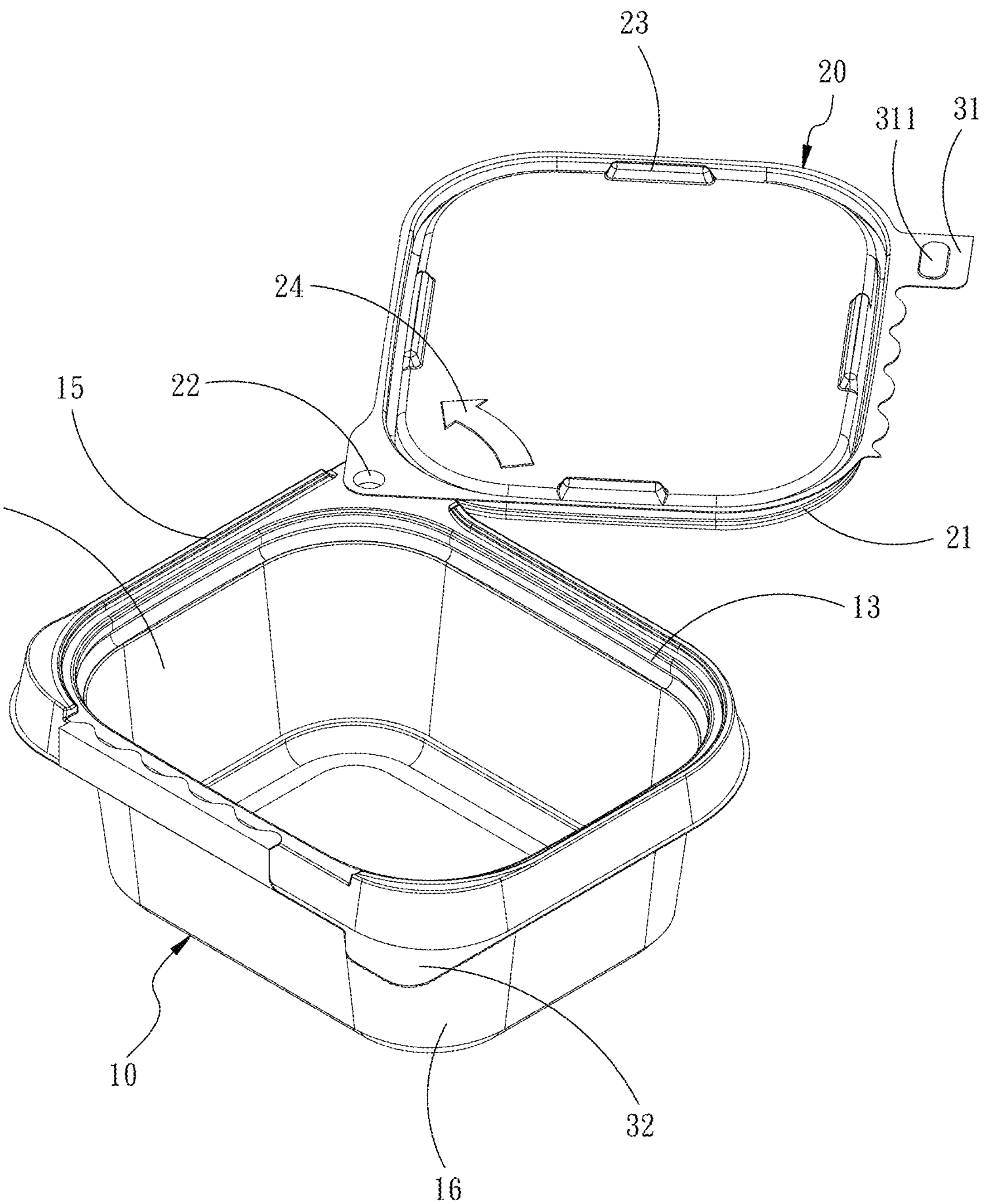


FIG. 11

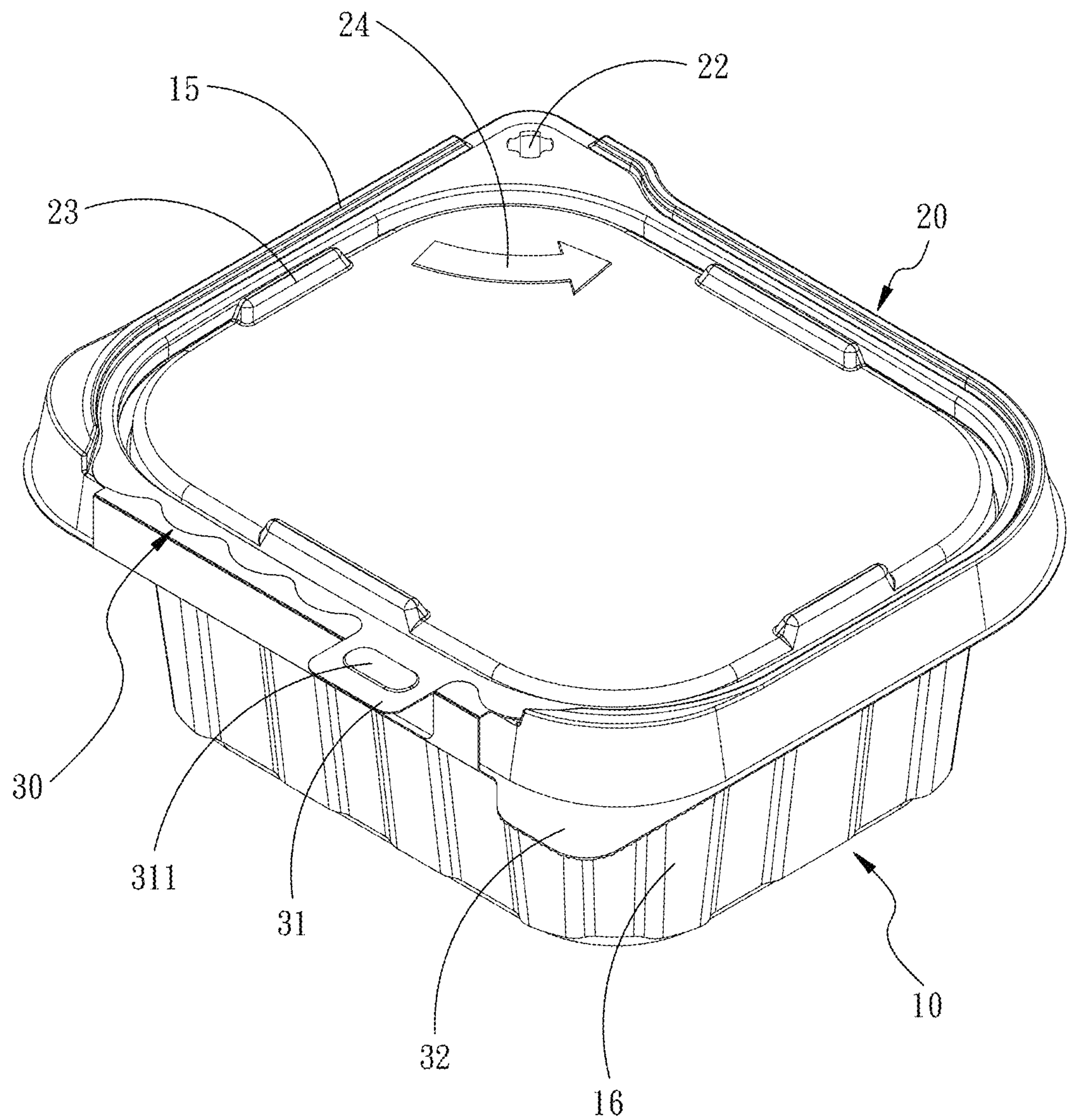


FIG. 12

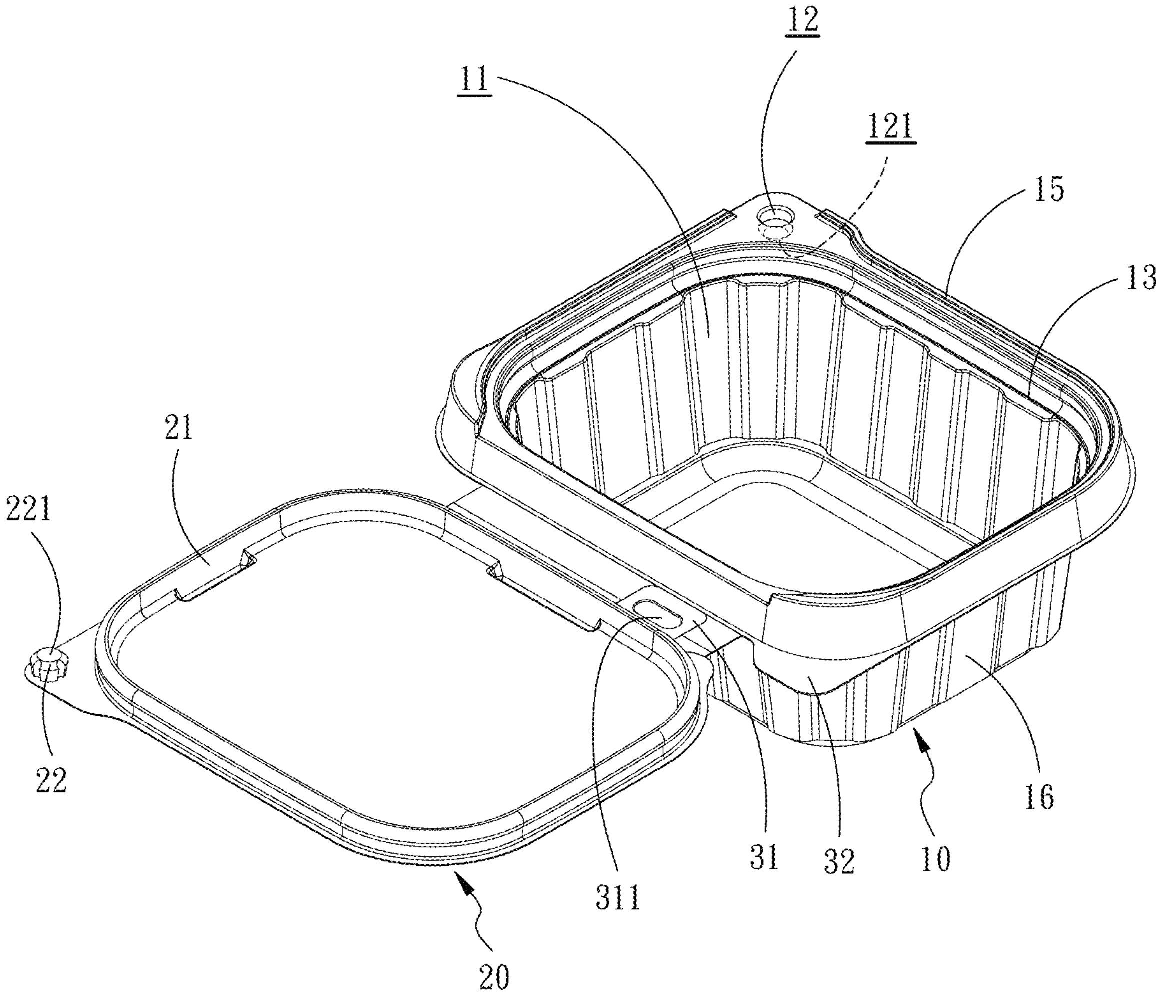


FIG. 13

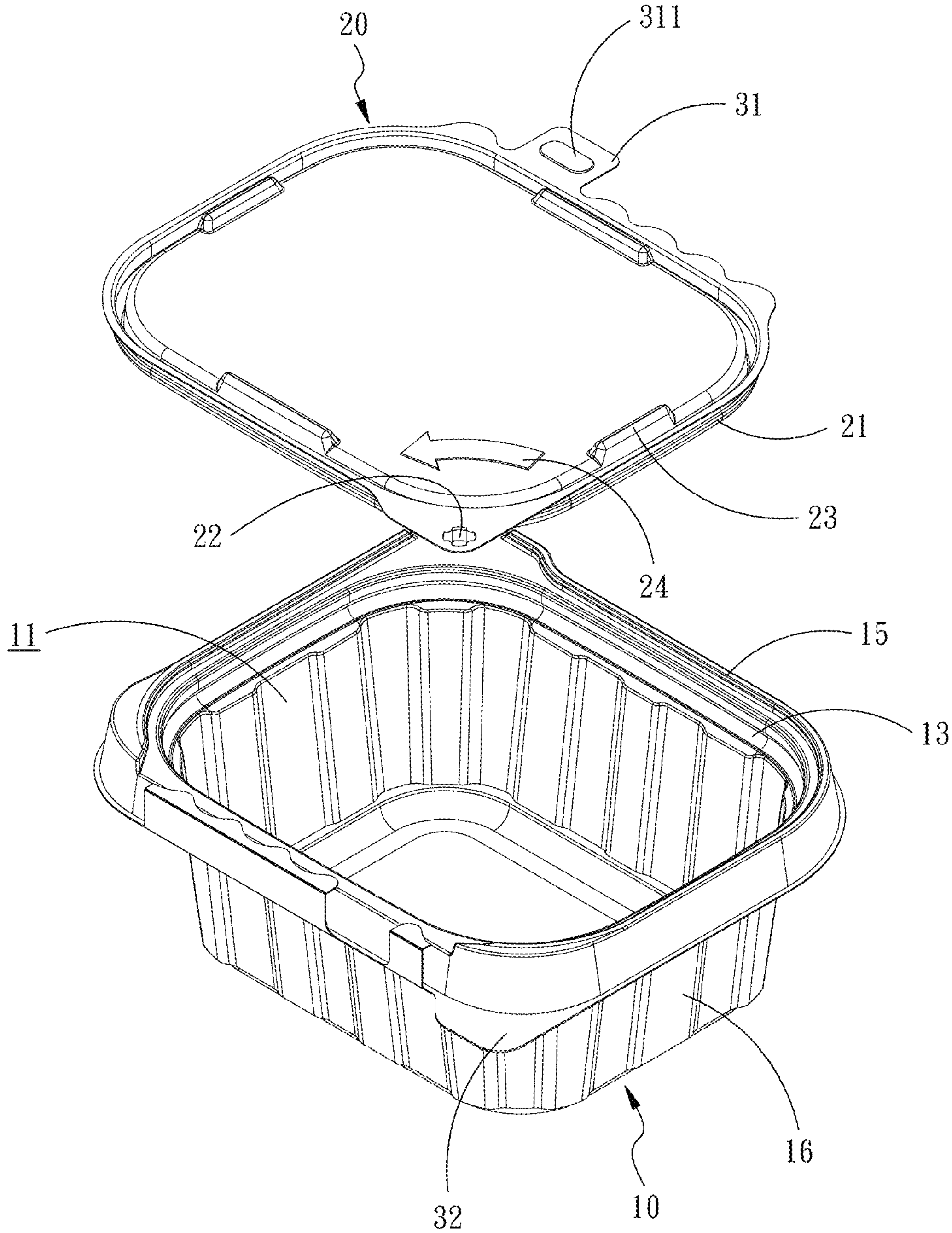


FIG. 14

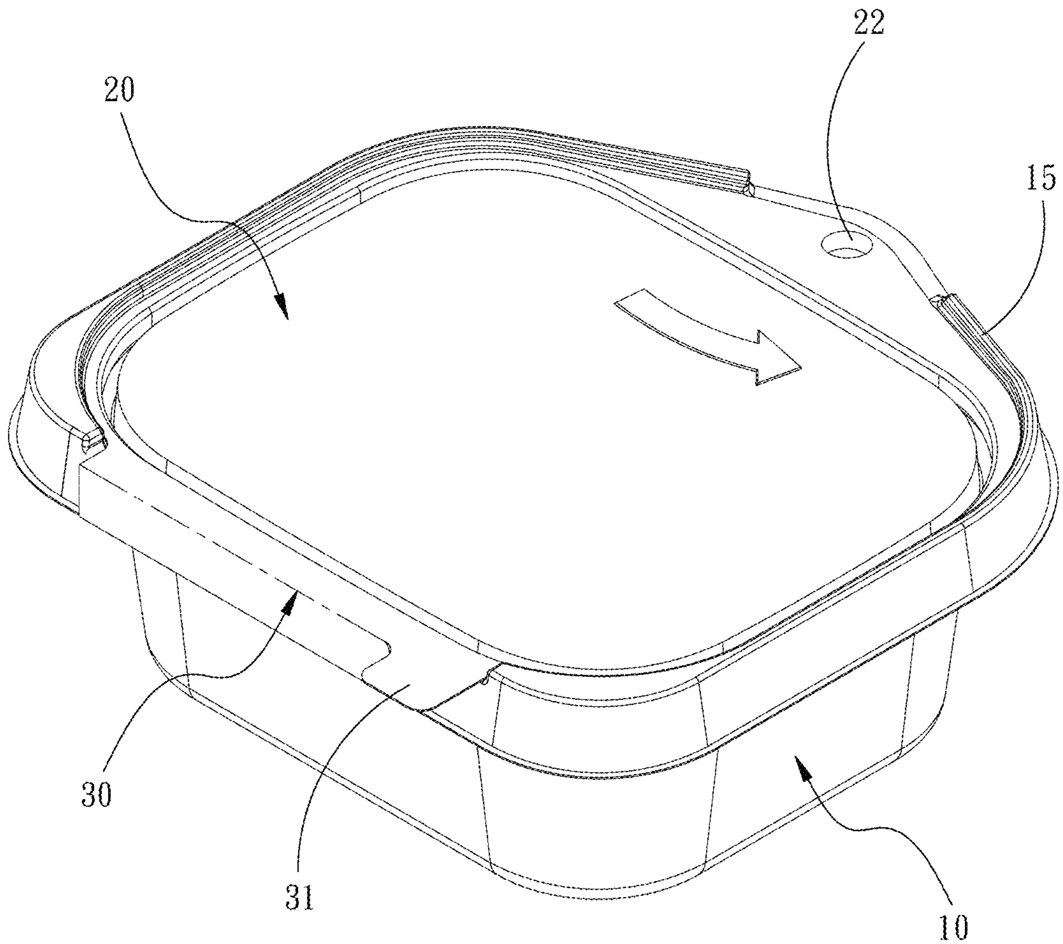


FIG. 15

TAMPER-EVIDENT CONTAINER WITH ROTATING LID

BACKGROUND OF THE INVENTION

1. Technical Field

The present invention relates to containers, and more particularly to a tamper-evident container with a rotating lid.

2. Description of Related Art

The long-lasting pandemic changed how people eat and many former regular diners are now loyal customers of food delivery services. Consequently, even in the "after" phase of the pandemic, food delivery still plays a key role in the catering market.

With the surge of transactions, disputes about delivery services increase. For example, some consumers complained that delivery drivers opened and sneaked their orders. Such an accusation is however hard to be ascertained because many food containers are not sealed. Similar cases can also be seen in supermarkets/hypermarkets, as food products packaged with containers may be opened or broken stealthily by unauthorized people without being noted by people buy the products later.

These issues reflect some shortcomings of the existing food containers. First, if a delivered or sold product has previously been maliciously unsealed and/or contaminated, a consumer taking it without any precautions may be exposed to health hazards, such as sitotoxism, and this goes against consumer rights on intactness and safety of food. Moreover, since consumers' loyalty and trust to brands are always based on quality and safety of products, any food product that is doubtable in terms of quality and safety will eventually be spurned by consumers.

Additionally, most existing food containers are made as disposable one-piece meal boxes. Since the lid and the box body are simply y connected without having any pivot therebetween, the lid flipped up, after the flipping force disappears, will drop back toward the box body by gravity unless the user keeps pushing or uses an external object to hold the lid in the open position. Thus, use of the conventional container is far from convenient.

In view of this, improvement of the conventional food containers is desirable.

SUMMARY OF THE INVENTION

A primary objective of the present invention is to solve the issues of conventional food containers by protecting a container from being secretly unsealed and ensuring that the opened lid remains in the open position to allow easy access to the food in the opened container.

To achieve the foregoing objective, one embodiment of the present invention provides a tamper-evident container with a rotating lid. The container comprises a box body, the lid, and a pre-scored connecting portion. The box body defines therein an accommodating space. The box body has a horizontal end surface provided with a first locking portion. The accommodating space has a first combining unit close to the horizontal end surface of the box body. The lid has a bottom surface provided with a second combining unit and a second locking portion. The second locking portion is locationally corresponding to the first locking portion and is for being engaged with the first locking portion. The second combining unit is to be detachably fit into the first combining unit so as to seal the accommodating space. The pre-scored connecting portion connects between the box body and the lid. The pre-scored connecting portion is located remote from the first locking portion and the second locking portion. When the pre-scored connecting portion is broken and the second combining unit and the first combining unit are separated from each other, the second locking portion is allowed to be horizontally rotated with respect to the first locking portion.

Thereby, after food is placed into the box body and the lid is combined with the box body, the tamper-evident container is sealed for safe delivery. A user receiving the delivery can only open the container by removing the pre-scored connecting portion between the lid and the box body and then rotating the lid away from the box body. Since the lid is horizontally rotated away from the box body, it will not fall back to the box body due to the gravity. In addition, by rotating the first locking portion with respect to the second locking portion to open and close the container, the container can be reused.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of a container according to a first embodiment of the present invention, wherein the lid has been opened.

FIG. **2** is another perspective view of the container of FIG. **1**, wherein the lid has been closed.

FIG. **3** is a cross-sectional view of the container of FIG. **1**, taken along Line A-A of FIG. **2**.

FIG. **4** depicts the container in FIG. **2** from a different angle of view.

FIG. **5** is another perspective view of the container of FIG. **1** showing that the lid has been detached from the box body.

FIG. **6** is another perspective view of the container of FIG. **1** showing that the lid has been rotated away from the box body.

FIG. **7** is a perspective view of a container according to a second embodiment of the present invention, wherein the lid has been closed.

FIG. **8** is a cross-sectional view of a first aspect of the present invention taken along Line B-B of FIG. **7**.

FIG. **9** is a cross-sectional view of a second aspect of the present invention taken along Line B-B of FIG. **7**.

FIG. **10** is a partial, zoomed-in view of FIG. **9**.

FIG. **11** is another perspective view of the container of FIG. **7**, wherein the lid has been rotated away from the box body.

FIG. **12** is a perspective view of a container according to a third embodiment of the present invention, wherein the lid has been closed.

FIG. **13** is another perspective view of the container of FIG. **12**, wherein the lid has been opened.

FIG. **14** is another perspective view of the container of FIG. **12**, wherein the lid has been rotated away from the box body.

FIG. **15** is a perspective view of a container according to a fourth embodiment of the present invention, wherein the second locking portion is at the middle of an edge of the lid.

DETAILED DESCRIPTION OF THE INVENTION

For further illustrating the means and functions by which the present invention achieves the certain objectives, the following description, in conjunction with the accompanying drawings and preferred embodiments, is set forth as below to illustrate the implement, structure, features and effects of the subject matter of the present invention.

Referring to FIG. **1** through FIG. **15**, the present invention provides a tamper-evident container with a rotating lid, comprising a box body **10**, a lid **20**, and a pre-scored connecting portion **30**. The box body **10** defines an accommodating space **11**. The box body **10** has its horizontal end surface provided with a first locking portion **12**. The first locking portion **12** in the present embodiment is a column-like socket and has an inner diameter that is constant or is gradually enlarged as the first locking portion **12** extends away from its free end. A first combining unit **13** is provided in the accommodating space **11** and close to the horizontal end surface of the box body **10**. The first combining unit **13** in the present embodiment is a stepped recess. The lid **20** has its bottom surface provided with a second combining unit **21** and a second locking portion **22**. The second locking portion **22** in the present embodiment is formed as a column-like boss and has an outer diameter that is constant or is gradually enlarged as the second locking portion **22** extends toward its free end. The second locking portion **22** is locationally corresponding to the first locking portion **12** so as to be fittingly combined with the first locking portion **12** and thereby firmly hold the box body **10** and the lid **20** together. In addition, if the second locking portion **22** is tapered away from its free end, and the first locking portion **12** is tapered toward its free end, it would be difficult to combine the second locking portion **22** with the first locking portion **12**. Moreover, after the second locking portion **22** and the first locking portion **12** are combined, it would be even more difficult to pull the second locking portion **22** out of the first locking portion **12**. With this configuration, a tricky attempt to forcedly push the second locking portion **22** out by applying a force to the bottom of the first locking portion **12** is unlikely to succeed. The second combining unit **21** in the present embodiment is a raised circular portion. The second combining unit **21** is to be detachably fit into the first combining unit **13** in the form of the stepped recess, so as to thereby seal the accommodating space **11**. The pre-scored connecting portion **30** is connected between the box body **10** and the lid **20**, and is located at the side remote from the first locking portion **12** and the second locking portion **22**. When the pre-scored connecting portion **30** breaks and separation between the second combining unit **21** and the first combining unit **13** happens, the second locking portion **22** can be horizontally rotated against the first locking portion **12**.

Thereby, when the lid **20** is closed to the box body **10**, and the first combining unit **13** is combined with the second combining unit **21**, the box body **10** is sealed. Combination between the first locking portion **12** and the second locking portion **22** effectively prevents that the lid **20** and the box body **10** separate from each other. In virtue of the almost impregnable combination between the first locking portion **12** and the second locking portion **22**, the only practical way to open the sealed container is to break the pre-scored connecting portion **30** and pull the lid **20** away from the body box **10** upward from the broken pre-scored connecting portion **30** until the first combining unit **13** and the second combining unit **21** separate from each other. At this time, the lid **20** can be horizontally rotated to expose the accommodating space **11**.

Furthermore, since the lid **20** is horizontally rotated away from the box body **10**, opposite to the prior-art cover that can easily drop by gravity and needs additional positioning means, the lid **20** can stay at the open position without hindering the user from accessing food in the accommodating space **11**.

Still referring to FIG. **1**, the free end of the second locking portion **22** is formed with a tapered end **221** and the first locking portion **12** has a tapered recess **121** at its bottom edge. After the tapered end **221** is fit in the tapered recess **121**, the first locking portion **12** and the second locking portion **22** are firmly combined.

Referring to FIG. **2**, a prompt unit **24** is provided at the surface of the lid **20** to prompt the user the recommended direction for rotating the lid **20** to expose the accommodating space **11** of the box body **10** after the pre-scored connecting portion **30** is broken.

Still referring to FIG. **2**, a retaining ridge **15** is provided at the top edge of the box body **10** without contacting the first locking portion **12**. When the lid **20** is closed to the box body **10**, the retaining ridge **15** is higher than the periphery of the lid **20**. Thereby, when the lid **20** is closed to the box body **10**, it would be difficult for a person to open the container by trying to pry the lid **20** at its periphery.

Referring to FIG. **2** through FIG. **4**, and also FIG. **1**, in one embodiment, the pre-scored connecting portion **30** has its one end formed with a tab **31**. When the lid **20** is closed to the box body **10**, the tab **31** cocks up besides the lid **20**. The tab **31** has a curved periphery. The tab **31** serves to facilitate tearing up the pre-scored connecting portion **30**. Moreover, by grabbing the tab **31**, a user can pull up the lid **20** easily to separate the first combining unit **13** from the second combining unit **21**. Therein, the tab **31** may be alternatively provided at an arbitrary position along the lateral of the lid **20** and its arrangement is not limited to the ends or the center of the lid **20**.

Referring now to FIG. **2** and FIG. **7**, the pre-scored connecting portion **30** may be made to be straight or wavy. As shown in FIG. **7**, when it is made as a wavy line, it may have a smooth profile without any acute angles. With such a design, the risk that the torn pre-scored connecting portion **30** cuts the user can be eliminated, thereby improving use safety.

Instead of being located at the corner of the box body **10** remote from the pre-scored connecting portion **30** as shown in FIG. **1**, the first locking portion **12** in another embodiment may alternatively be located at the middle of the edge of the box body **10** remote from the pre-scored connecting portion **30** (as shown in FIG. **15**), so that the lid **20** and the box body **10** are connected and can be rotated with respect to each other in a symmetrical and balanced manner.

Now referring to FIG. **7** through FIG. **11**, in the second embodiment, the lid **20** has its top surface formed with a plurality of retaining bars **23**. These retaining bars **23** are to be engaged with the bottom surface of the box body **10** of another container of the present invention so as to hold the box body **10** in position firmly. With this configuration, plural containers of the present invention stacked vertically can be handled and/or stored stably for safe transportation and easy storage.

Referring to FIG. **9** and FIG. **10**, the box body **10** is provided with a plurality of retaining portions **231** locationally corresponding to the retaining bars **23** of the lid **20**. The retaining portions **231** are to be engaged with the retaining bars **23** of the lid **20**, so as to strengthen tightness between the lid **20** and the box body **10**, and enhance the sealing performance of the container.

Referring to FIG. **12** through FIG. **14**, in the third embodiment of the present invention, the second locking portion **22** is formed as a boss that has a cross-shaped section, and the first locking portion **12** is formed as a correspondingly shaped socket. With the cross-shaped section, the second locking portion **22** has improved structural resilience, making it easier to combine the second locking portion **22** and the first locking portion **12**. Additionally, the cross-shaped boss of the second locking portion **22** may be tapered away from the free end, and the first locking portion **12** is tapered toward the free end. While these configurations add difficulty in combining the second locking portion **22** and the first locking portion **12**, they can effectively secure the combination by preventing the second locking portion **22** from being pulled out of the first locking portion **12**. Even a tricky attempt to forcedly push the second locking portion **22** out by applying a force to the bottom of the first locking portion **12** is unlikely to succeed. In other words, the second locking portion **22** formed as the cross-shaped boss is easier to combine with the first locking portion **12** as compared to the second locking portion **22** of the previous embodiment formed as a column-like boss, yet the resulting combination of the first locking portion **12** and the second locking portion **22** achieves the comparable tightness and security, thereby providing a preferrable effect of combination and tamper-evidence.

Still referring to FIG. **12** through FIG. **14**, the box body **10** is further provided with a lug **32** that is close to the tab **31**. The lug **32** facilitates force application. The tab **31** is provided with a force application portion **311** for providing frictional resistance. As shown, the force application portion **311** has its side facing the tab **31** formed as a bulged surface and has the opposite side formed as a recessed surface, so as to ensure good friction for a user to easily apply a pulling force when pulling the tab **31**. Alternatively, the force application portion **311** may be provided with grooves or nodules to similarly provide frictional resistance. With these configurations, a user can hold the lug **32** with one hand and grab and pull the tab **31** with the other hand to remove the pre-scored connecting portion **30**.

FIG. **15** depicts the fourth embodiment of the present invention. Therein, the second locking portion **22** is located at the middle of an edge of the lid **20**. Although this is not shown, the first locking portion **12** is located on the box body **10** to be locationally corresponding to the second locking portion **22**.

In one embodiment, the box body **10** has its lateral walls each provided with reinforcing ribs **16** that are formed by folding the material of the box body **10**. The reinforcing ribs **16** enable the box body **10** to have increased structural strength without increasing its thickness. In other words, with the same material weight and manufacturing costs, the rib-equipped box body **10** is less likely to collapse than its plain counterpart.

In virtue of the foregoing features, the present invention provides the following advantages:

1. Easy tamper detection: with provision of the pre-scored connecting portion **30**, any attempt to open the lid **20** will break the pre-scored connecting portion **30**, and this makes it impossible for an unauthorized person to open the lid **20** and access and/or tamper objects in the box body **10** secretly.

2. Convenient use: the lid **20** that has been opened through removal of the pre-scored connecting portion **30** and horizontal rotation with respect to the box body **10** can stay still in the open position, and this saves a user from the trouble of using additional means or efforts to make the lid **20** remain open.

3. Reliable combination: the tapered recess **121** and the tapered end **221** jointly secure the combination between the first locking portion **12** and the second locking portion **22**, and in turn guarantee reliable combination between the lid **20** and the box body **10**. Even when the lid **20** is being rotated, the lid **20** and the box body **10** can be well combined and are unlikely to depart from each other.

4. Difficulty in removing the lid **20**: with the second locking portion **22** formed as the cross-like boss that is tapered away from the free end, it is almost impossible to open the sealed container by detaching the second locking portion **22** from the first locking portion **12** without breaking the container.

5. Effortless removal of the pre-scored connecting portion **30**: by grabbing the tab **31** having the curved periphery, a user can pull up and remove the pre-scored connecting portion **30** safely. Additionally, the force application portion **311** optionally provided on the tab **31** makes it easy for the user to apply force and remove the pre-scored connecting portion **30** effortlessly.

6. Stackability: provision of the retaining bars **23** allows the container of the present invention to be stacked with good stability, thereby facilitating handling, transportation, and storage.

7. Good tightness: in addition to the combination between the first combining unit **13** and the second combining unit **21**, the matching retaining bars **23** and retaining portions **231** further secure the combination of the box body **10** and the lid **20**, so the closed container is more structurally stable and has good tightness and sealing performance.

8. High structural strength: the reinforcing ribs **16** at the lateral walls of the box body **10** enhance the structural strength of the box body **10**, thereby making the box body **10** more durable and better load-bearing.

The present invention has been described with reference to the preferred embodiments and it is understood that the embodiments are not intended to limit the scope of the present invention. Moreover, as the contents disclosed herein should be readily understood and can be implemented by a person skilled in the art, all equivalent changes or modifications which do not depart from the concept of the present invention should be encompassed by the appended claims.

What is claimed is:

1. A tamper-evident container with a rotating lid, comprising:

a box body, defining therein an accommodating space, and having a horizontal end surface provided with a first locking portion, wherein a first combining unit is provided in the accommodating space close to the horizontal end surface of the box body;

the lid, having a bottom surface provided with a second combining unit and a second locking portion, wherein the second locking portion is locationally corresponding to the first locking portion and configured to be engaged with the first locking portion, and the second combining unit is configured to be detachably fit into the first combining unit so as to seal the accommodating space; and

a pre-scored connecting portion, connecting between the box body and the lid and being located remote from the first locking portion and the second locking portion;

whereby when the pre-scored connecting portion is broken and the second combining unit and the first combining unit are separated from each other, the second locking portion is allowed to be horizontally rotated with respect to the first locking portion.

2. The tamper-evident container of claim **1**, wherein the second locking portion is formed as a column-like boss, and the first locking portion is formed as a column-like socket matching the second locking portion.

3. The tamper-evident container of claim **1**, wherein the second locking portion is formed as a cross-shaped boss, and the first locking portion is formed as a cross-shaped socket matching the second locking portion.

4. The tamper-evident container of claim **1**, wherein the second locking portion is tapered away from a free end thereof, and the first locking portion is tapered toward a free end thereof.

5. The tamper-evident container of claim **4**, wherein the free end of the second locking portion is formed with a tapered end, and the first locking portion has a bottom edge correspondingly provided with a tapered recess.

6. The tamper-evident container of claim **5**, wherein the first locking portion is located at a corner of the box body remote from the pre-scored connecting portion.

7. The tamper-evident container of claim **5**, wherein the first locking portion is located at a middle of an edge of the box body remote from the pre-scored connecting portion.

8. The tamper-evident container of claim **5**, wherein the pre-scored connecting portion has one end provided with a tab, so that when the lid is closed to the box body, the tab cocks up besides the lid.

9. The tamper-evident container of claim **8**, wherein the tab has a curved periphery.

10. The tamper-evident container of claim **9**, wherein the tab is provided with a force application portion for providing frictional resistance.

11. The tamper-evident container of claim **10**, further comprising a retaining ridge that is provided at a top edge of the box body without contacting the first locking portion, so that when the lid is closed to the box body, the retaining ridge is higher than a periphery of the lid.

12. The tamper-evident container of claim **11**, further comprising a plurality of retaining bars provided at a top surface of the lid for retaining a bottom surface of another said box body.

13. The tamper-evident container of claim **12**, wherein the box body comprises a plurality of retaining portions locationally corresponding to the retaining bars of the lid, and the retaining portions are configured to be engaged with the retaining bars of the lid.

14. The tamper-evident container of claim **13**, further comprising reinforcing ribs provided at lateral walls of the box body.

15. The tamper-evident container of claim **14**, wherein the pre-scored connecting portion is made to be straight or wavy.

16. The tamper-evident container of claim **15**, further comprising a prompt unit provided at the top surface of the lid.

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